

# LITERATURE REVIEW & RESEARCH GAP TABLE

New India Healthcare Data 2026 — comparative review of the three user-supplied sources and the uploaded healthcare dataset

| S. No. | Author(s) & Year            | Title / Work                                                                                                                                                                                             | Dataset / Data Source                                                                                                                                                    | Method / Focus                                                                                                                                                                                                           | Key Features / Input                                                                                                                                                  | Results / Contribution                                                                                                                                                                                                                                                  | Limitations                                                                                                                                                                                                                                                           | Research Gap vs. <i>New India Healthcare Data 2026</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | IEEE Xplore 2025/2026*      | IEEE Xplore source<br><b>Document 10961906</b>                                                                                                                                                           | Dataset details could not be independently retrieved from IEEE Xplore because the linked record blocked automated access.                                                | Not safely reported without verified bibliographic metadata.                                                                                                                                                             | Not safely reported without verified bibliographic metadata.                                                                                                          | Not safely reported without verified bibliographic metadata.                                                                                                                                                                                                            | Primary source metadata was inaccessible in the research environment; reporting a title, sample size, algorithm or accuracy would risk inventing facts.                                                                                                               | <b>Gap to validate:</b> once the exact paper/dataset is confirmed, compare its population size, Indian coverage, disease scope, longitudinal span, clinical variables and model task against this project. The uploaded dataset is substantially structured for broad, multi-disease tabular healthcare analytics.                                                                                                                                                                                                                                                                                                                                     |
| 2      | Manish Kumar et al. 2026*   | <b>Integrated Multi-Task Explainable AI for Healthcare Prediction Using Heterogeneous Indian Patient Data</b><br><i>Linked through <i>Next-Generation Oncology: Nanomedicine and AI Applications</i></i> | Heterogeneous Indian patient data; exact sample size/variable list is not exposed in the accessible Google Books preview.                                                | Multi-task healthcare prediction with an <b>explainable AI (XAI)</b> orientation; exact architecture should be verified from the full chapter.                                                                           | Heterogeneous patient information; intended to support multiple healthcare prediction tasks and explanation.                                                          | Contribution is the integration of heterogeneous Indian patient information with multi-task prediction and explainability; no verified accuracy is stated here because the accessible preview does not expose the full experimental tables.                             | Dataset composition, sample size, target definitions and external validation cannot be confirmed from the accessible preview.                                                                                                                                         | <b>Major gap:</b> your dataset is a <b>100,000-record, 39-variable</b> structured healthcare table spanning <b>10 Indian states, 10 diagnosis groups and 2024–2026 visits</b> . It combines demographics, socioeconomic status, occupation, symptoms, glucose/HbA1c/cholesterol, BMI, diagnosis, treatment/outcome, imaging metadata, hospital type and insurance. A strong project contribution is to benchmark multiple ML models across several prediction targets and add SHAP/XAI, subgroup fairness and temporal/geographic validation. Unlike raw multimodal data, your imaging fields are metadata/categories rather than actual image pixels. |
| 3      | Singh et al. 2026           | <b>Addressing Bias, Privacy, Security, and Patient Autonomy in Artificial Intelligence (AI)-Driven Healthcare: A Review of Current Guidelines</b>                                                        | Narrative review; sources include legal/regulatory instruments, ethics frameworks, empirical studies and healthcare-AI case evidence rather than one predictive dataset. | Policy/ethical review of AI governance, privacy, security, bias, consent, transparency and accountability.                                                                                                               | Healthcare AI data; emphasis on privacy, patient autonomy, algorithmic bias, security, consent, governance and Indian DPDP context.                                   | The review concludes that responsible healthcare AI requires stronger governance, privacy/security-by-design, bias mitigation, meaningful consent and human oversight; it recommends privacy-preserving approaches such as federated learning and differential privacy. | It is policy-oriented and does not provide a single empirical predictive benchmark for a 100,000-record Indian tabular dataset.                                                                                                                                       | <b>Major gap:</b> your project can convert these governance recommendations into measurable analysis: test model performance across <b>gender, socioeconomic status, state and age groups</b> ; assess calibration/error differences; use explainability; document de-identification; and evaluate whether temporal/geographic shifts affect predictions. This connects <b>predictive performance + responsible AI</b> rather than treating them separately.                                                                                                                                                                                           |
| —      | <b>Your project</b><br>2026 | <b>New India Healthcare Data 2026</b>                                                                                                                                                                    | <b>100,000 patient records × 39 variables</b> ; 10 Indian states; visits from 2024–2026; 10 primary diagnosis groups; no missing values observed in the workbook.        | <b>Recommended:</b> supervised ML for diagnosis/risk/outcome prediction; baseline Logistic Regression, Decision Tree, Random Forest, Gradient Boosting/XGBoost; cross-validation + held-out test set; SHAP/LIME for XAI. | Demographic + socioeconomic + occupational + symptom + laboratory + BMI + diagnosis + treatment + outcome + imaging metadata + hospital + insurance + time variables. | This is the <b>benchmarking opportunity</b> : move from single-disease or policy-focused work toward a broad Indian tabular healthcare prediction framework with multiple targets and transparent evaluation.                                                           | It is a structured project dataset rather than raw EHR, medical images, genomic/omics or free-text clinical notes. Patient-level leakage, synthetic-data provenance and representativeness should be explicitly documented before claiming clinical generalizability. | <b>Core research gap:</b> build a reproducible, explainable and fairness-aware ML framework for heterogeneous Indian healthcare data that evaluates <b>accuracy, precision, recall, F1, ROC-AUC, calibration, subgroup performance and temporal/geographic robustness</b> —while following privacy-by-design principles.                                                                                                                                                                                                                                                                                                                               |

**Dataset profile used for the comparison:** 100,000 records, 39 variables, 10 Indian states, 10 primary diagnosis categories, visit dates spanning 2024–2026. Variables cover demographics, socioeconomic/occupation, symptoms, fasting glucose, HbA1c, cholesterol, BMI, diagnosis, treatment/outcome, imaging metadata, hospital type, insurance and time features.

**Important source-verification note:** The IEEE Xplore record (Document 10961906) could not be retrieved because IEEE Xplore returned an access error to the research browser. The Google Books link exposed the book metadata but not the full chapter dataset/method tables. Therefore, this table deliberately avoids inventing bibliographic, dataset-size or accuracy values that could not be verified. The Cureus source was fully accessible and its governance recommendations were reviewed directly.

**Recommended thesis-level research gap statement:** Existing healthcare-AI literature often focuses on a specific prediction task, a narrow dataset, or governance/ethics in isolation. The New India Healthcare Data 2026 project can address the combined gap by developing and evaluating an explainable, fairness-aware, multi-target machine-learning framework on large-scale heterogeneous Indian tabular healthcare data, with subgroup, temporal and geographic robustness checks.

**Sources:** IEEE Xplore document 10961906; Google Books-linked work in *Next-Generation Oncology: Nanomedicine and AI Applications*; Singh et al., Cureus 18(2), e103999, DOI 10.7759/cureus.103999. Web sources accessed August 27, 2026.